

# SHREEYAN RIJAL

Computational Materials Physicist / Prospective PhD Researcher — Photovoltaic & Energy Materials

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## Research Profile

Computational materials physicist specializing in first-principles (density functional theory) modeling of photovoltaic and energy-conversion materials. Combines electronic-structure calculations, spin-polarized transport analysis, and device-level simulation to investigate stability, efficiency, and optoelectronic behavior in perovskite solar cells, doped semiconductors, and van der Waals heterostructures. Author of three first-author peer-reviewed publications, with a demonstrated ability to move independently from first-principles theory through simulation to publishable, physically interpretable results. Seeks to extend this work toward a doctoral research program developing computational and data-driven methods for next-generation solar cell and quantum/energy materials.

## Education

**B.Sc. Physics — St. Xavier's College, Tribhuvan University, Kathmandu, Nepal** **2021–2025**

*Coursework: Quantum Mechanics • Condensed Matter Physics • Electrodynamics • Thermodynamics • Computational Methods*

## Research Experience

**Independent Researcher, Computational Materials Physics — St. Xavier's College**

- Performed density-functional-theory (DFT) modeling in Quantum ESPRESSO and device-level simulation in OghmaNano to characterize perovskite and thin-film solar cell architectures under thermal and optical stress, identifying parameter regimes that improve stability and efficiency.
- Executed spin-polarized first-principles calculations of Mn-doped silicon, resolving impurity-induced near-metallicity and enhanced infrared optical activity with direct relevance to spintronic device design.
- Modeled strain-engineered optoelectronic and thermoelectric properties of van der Waals heterostructures (MoSSe/Ti<sub>2</sub>CO<sub>2</sub>, Hf<sub>2</sub>CO<sub>2</sub>/blue phosphorus, Zr<sub>2</sub>CO<sub>2</sub>/blue phosphorus) using first-principles electronic-structure methods.
- Built and applied Python-based scientific computing workflows for data analysis and machine-learning-assisted interpretation of simulation and experimental datasets.
- Collaborated across multi-institutional research teams, translating first-principles results into three first-author peer-reviewed publications and several manuscripts in active preparation.

## Publications (Peer-Reviewed, First Author)

1. **Rijal, S.**, Sah, N., Chaudhary, P., & Jha, V.K. (2026). Enhancing the stability and efficiency of perovskite solar cells by optimization of electrical parameter under thermal and optical stress; *Optical and Quantum Electronics* (Springer), 58, 288; doi.org/10.1007/s11082-026-08876-3
2. **Rijal, S.**, Jha, V.K., et al. (2026). Impurity-induced near-metallicity, spin-polarized transport, and enhanced infrared optical activity in Mn-doped silicon: A spin-polarized first-principles study; *Physica B: Condensed Matter*, 739, 418967; doi.org/10.1016/j.physb.2026.418967
3. **Rijal, S.**, Chaudhary, P., Sharma, S., Thada Magar, M.T., Regmi, B., Adhikari, B. et al. (2026). Time-frequency characterization of prompt penetration electric fields across distinct geomagnetic regimes; *Advances in Space Research* (In Press); doi.org/10.1016/j.asr.2026.06.060

## Ongoing Research Projects

- Strain-Engineered Optoelectronic and Thermoelectric Properties of MoSSe/Ti<sub>2</sub>CO<sub>2</sub> van der Waals Heterostructure: A First-Principles Study — *Journal of Physics and Chemistry of Solids* (Lead Author)
- Strain-Engineered Optoelectronic and Thermoelectric Properties of Hf<sub>2</sub>CO<sub>2</sub>/Blue Phosphorus and Zr<sub>2</sub>CO<sub>2</sub>/Blue Phosphorus van der Waals Heterostructures: First-Principles Studies (Lead Author, two manuscripts)
- Double Perovskite Energy Material Study — *Advanced Energy Materials* (Co-Author)

## Technical Skills

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**DFT & Simulation:** Quantum ESPRESSO, Gaussian, OghmaNano (device simulation), ORCA (collaborator)  
**Programming & Analysis:** Python (scientific computing, machine learning, automation), MATLAB, C, LaTeX  
**Physics Areas:** Condensed matter and semiconductor physics, electronic structure, spintronics, photovoltaics, optoelectronics, thermoelectrics

## Leadership & Mentorship

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- **Library Council Representative**, Physics Department, St. Xavier's College — Sustained institutional leadership and service throughout all four years of undergraduate study.
- **Research Mentor**, St. Xavier's College — Guide junior undergraduates into DFT research and scientific writing.
- **LSRA Mentor** — Mentor a candidate for the Fathers Locke & Stiller Research Award.

## Languages

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Nepali (Native) • English (Fluent, Professional Proficiency) • Hindi (Fluent, Professional Proficiency)

## References

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**Dr. Binod Adhikari** — Supervisor; Former Head of Department, St. Xavier's College  
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